CS 310 : Automata Theory

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How to use pumping lemma?

Recall : Pumping Lemma as an Implication

L is regular $\Rightarrow$

$$\exists p \forall w \in L. [|w| \geq p \Rightarrow$$
$$\exists x, y, z (w = xyz \wedge y \neq \epsilon \wedge |xy| \leq p \wedge (\forall i \geq 0, xy^i z \in L))]$$

Recall : Its contrapositive

$$\forall p \, \exists w \in L. \big[|w| \geq p \wedge \ \forall x, y, z \big((w = xyz \wedge y \neq \epsilon \wedge |xy| \leq p)$$
$$\Rightarrow \exists i \geq 0. \ xy^i z \notin L\big)\big]$$
$$\Rightarrow L \text{ is not regular}$$

Contrapositive of pumping lemma

In words,

Theorem

If

- for all p,
- there is a word $w \in L$ such that $|w| \geq p$,
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We have proven that L is not regular.

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- Now we choose $k = 0$ for each i and j. The corresponding word is
$$0^i(0^j)^0 0^{p-j-i}1^n = 0^{p-j}1^p.$$
- Clearly, $0^{p-j}1^p \notin L$. Therefore, L is not regular.

Example 3: using pumping lemma

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- Let $k = 2$ for each i and j.

$$0^i(0^j)^2 0^{p-j-i}110^p = 0^{p+j}110^p \notin L_{rev}$$

Properties of regular languages

Exercises:

1. Is subset of any regular language regular?
2. Is superset of any regular language regular?
3. If $L_1 \cup L_2$ is regular, then are L_1 and L_2 regular?
4. Is this regular: $\{0^n 1^m | n \neq m\}$?
5. Is this regular: $\{0^n 1^m | n > m\}$?
6. Is this regular: $\{0^n 1^m \mid n \text{ is odd and } m \text{ is even } \}$
7. Is this regular: $\{0^n 1^{n+1} | n \geq 0\}$

Example 4 : not using pumping lemma

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- Suppose L is regular. Note that $\Sigma = \{0, 1\}$.
- Consider the language $L' = \overline{L} \cap 0^* 1^*$, where $\overline{L}$ is the complement of L.
- This must be regular by closure properties.
- But $L' = \{0^n 1^n \mid n \in \mathbb{N}\}$, which we know is not regular.
- Hence we have a contradiction and L is not regular.

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- if $k \geq 2$, we get $xy^kz = 0^i 0^{j+2} 0^{p+1-j-i} 1^p \in L$, so NOT enough!

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- This completes the proof.

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- Let $k = n - j$. Therefore, $|xy^kz| = (n - j)(1 + j)$.
- Since both $(n - j) > 1_{\text{(why?)}}$and $1 + j > 1_{\text{(why?)}}$, $xy^kz \notin L_{prime}$.

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  Therefore, $xy^2 z \notin L$.

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Pumping Lemma and Non-Regularity

Pumping Lemma is a necessary but not sufficient condition for regularity.

$F = \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } (i = 1 \Rightarrow j = k)\}$

- F is **not** regular
- F **satisfies** the pumping lemma

$F = \{a^i b^j c^k \mid i, j, k \geq 0, (i = 1 \Rightarrow j = k)\}$ not regular

Suppose F were regular.

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So it suffices to show $L = \{ab^n c^n \mid n \geq 0\}$ is not regular

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Suppose it were regular. Let p be its pumping length.

- Take $w = ab^pc^p$, $|w| = 2p + 1 \geq p$.
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- Hence, x, y consist only of a's and/or b's.

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- Take $i = 0$: $xy^0z = xz$ removes the $|y| > 0$ symbols of y from the a/b-prefix, while leaving all p c's untouched.

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- So xz has fewer than p a's and b's combined (or lost its single a) but has exactly p c's $\Rightarrow xz \notin L$.

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So L, and hence F, is not regular

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$F = \{a^i b^j c^k | i, j, k \geq 0, (i=1 \Rightarrow j=k)\}$

F's only constraint is at $i = 1$. For $i \neq 1$, j, k can be anything.

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F's only constraint is at $i = 1$. For $i \neq 1$, j, k can be anything.

Strategy

Pump only inside the a's, chosen so the count of a is not 1

Case analysis: $i = 0$ and $i = 1$

Let $w = a^i b^j c^k \in F$, $|w| \geq 2$.

- $i = 0$: take $x = \varepsilon$, $y =$ first symbol of w, $z =$ rest.

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  Pumping only changes the b or c count; $i \neq 1 \Rightarrow$ is in F.

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- $i = 1$: then $j = k$. Take $x = \varepsilon$, $y = a$, $z = b^j c^k$. $xy^\ell z = a^\ell b^j c^k$.
  - At $\ell = 1$: in F as $j = k$
  - For $\ell \neq 1$: a count $\neq 1 \Rightarrow$ is in F.

Case analysis: $i = 2$ and $i \geq 3$

Let $w = a^i b^j c^k \in F$, $|w| \geq 2$.

- $i = 2$: take $x = \varepsilon$, $y = aa$, $z = b^j c^k$.

Case analysis: $i = 2$ and $i \geq 3$

Let $w = a^i b^j c^k \in F$, $|w| \geq 2$.

- $i = 2$: take $x = \varepsilon$, $y = aa$, $z = b^j c^k$.

  $xy^\ell z = a^{2\ell} b^j c^k$. The a-count 2ℓ is always **even** $\Rightarrow \neq 1 \Rightarrow$ is in F

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- $i \geq 3$: take $x = a$, $y = a$, $z = a^{i-2} b^j c^k$ (note $|xy| = 2 \leq p$).

Case analysis: $i = 2$ and $i \geq 3$

Let $w = a^i b^j c^k \in F$, $|w| \geq 2$.

- $i = 2$: take $x = \varepsilon$, $y = aa$, $z = b^j c^k$.

  $xy^\ell z = a^{2\ell} b^j c^k$. The a-count 2ℓ is always **even** $\Rightarrow \neq 1 \Rightarrow$ is in F

- $i \geq 3$: take $x = a$, $y = a$, $z = a^{i-2} b^j c^k$ (note $|xy| = 2 \leq p$).

  $xy^\ell z = a^{(i-1)+\ell} b^j c^k$.
  Since $i \geq 3$: a-count $= i-1+\ell \geq 2$ for all $\ell \geq 0 \Rightarrow \neq 1 \Rightarrow$ is in F.

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  Since $i \geq 3$: a-count $= i-1+\ell \geq 2$ for all $\ell \geq 0 \Rightarrow \neq 1 \Rightarrow$ is in F.

Every case yields a valid pumping decomposition. So F satisfies the pumping lemma, despite being non-regular.

Takeaway

What this example shows

The pumping lemma can only **disprove** regularity, it can never **prove** it, and it cannot **always** detect non-regularity either.

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To show F is non-regular, we used closure under intersection with a simpler regular language, reducing to the known non-regular $\{ab^nc^n \mid n \geq 0\}$.

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The pumping lemma can only **disprove** regularity, it can never **prove** it, and it cannot **always** detect non-regularity either.

To show F is non-regular, we used closure under intersection with a simpler regular language, reducing to the known non-regular $\{ab^nc^n \mid n \geq 0\}$.

If the pumping lemma seems to "work" for a language, that's not proof of regularity.

Where does the pumping lemma fall short?

We consider a split $w = xyz$ with $|xy| \leq p$ the pump window is anchored at a prefix, trying to find a bad word.

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- For F, *exactly one* a forces $j = k$ is at the front, used by the pump window
- the pumping lemma only needs one safe split to exist
- we can pick the pump a option, which is the safe $i \neq 1$ case,
- the unsafe option (pumping the b's) is never forced